Scalability & Future Plan

Date	3 October 2026
Team ID	SWTID-2026-5338
Project Name	ComicCraft-AI Comic Creator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current **ComicCraft – AI Comic Story Creator using Gemini Models** solution can be scaled to support more users, higher AI generation requests, increased storage requirements and additional features in the future. It also presents the roadmap for improving ComicCraft beyond its current local implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently runs locally and is not hosted online.	Users cannot access ComicCraft remotely through the internet.	High
2	AI story and image generation depend on external Gemini and Hugging Face API availability and usage limits.	Generation may be delayed or temporarily unavailable when API limits are reached.	High
3	Generated comic images and PDF files are currently stored locally.	Large numbers of generated comics may require more storage and management.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Application runs locally and is mainly designed for individual use.	Deploy the FastAPI application on a cloud server and use scalable infrastructure to support multiple users.
2	Data Storage	Generated images and PDF files are stored locally.	Use cloud storage for storing generated comic images and PDF files.
3	Performance	AI generation requests are processed through external APIs.	Introduce asynchronous processing, request queuing and optimized API handling for larger workloads.
4	Security	API keys are stored using `.env` configuration.	Use secure cloud environment variables or secret-management services when deploying the application online.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target	Expected Impact

		Timeline	
Phase 2	Cloud deployment of ComicCraft	Future Enhancement	Allows users to access the application online from different devices.
Phase 3	User accounts and cloud storage for generated comics	Future Enhancement	Allows users to save, manage and access their previously generated comics.
Phase 4	Advanced comic customization such as more panels, character consistency and additional art styles	Future Enhancement	Provides a richer and more flexible comic creation experience.